CancerBot: A Retrieval-Augmented Generation-based Cancer Chatbot Using Large Language Models

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***Abstract*— Cancer is the second leading cause of death globally, highlighting the urgent need for innovative solutions to support patients and healthcare providers. Remote and digital cancer care interventions, such as chatbots, are more relevant than ever, offering timely information and personalized assistance. With the rise of conversational AI, chatbots have emerged as a potential solution to bridge this gap, offering 24/7 assistance and personalized responses. This paper introduces CancerBot, a Retrieval-Augmented Generation (RAG) system designed to address the need for reliable, contextually relevant cancer-related information. Built on top of a large language model LLaMA-2, and integrated with a vector database, CancerBot retrieves relevant medical literature to generate accurate responses, minimizing the risk of hallucinations. Through synthetic testing and evaluation using metrics such as faithfulness, context relevancy, and response relevancy, our results demonstrate that the RAG-based CancerBot significantly outperforms the standard LLaMA-2 model in addressing cancer-specific queries. This advancement showcases the potential of RAG systems to improve patient care by providing reliable, context-driven information that enhances patient engagement and healthcare decision-making.**

Keywords*— *Cancer, Chatbot, Llama2, LLM, RAG, Vector Databases

I. Introduction

The World Health Organization (WHO) reports that cancer is the second leading cause of death globally. In 2022, there were approximately 20 million new cancer cases and 9.7 million cancer-related deaths worldwide. By 2040, these numbers are expected to rise significantly, with an estimated 29.9 million new cases and 15.3 million deaths annually [1]. This growing burden emphasizes the need for innovative solutions to enhance support for patients, caregivers, and healthcare providers. Timely diagnosis and optimal treatment are essential for improving cancer survival rates. Consequently, there is a rising demand for an effective healthcare system to address the complex treatment and care needs of cancer patients. Many patients experience challenges related to physical, emotional, and practical concerns, yet often struggle to find adequate information to manage these issues [2]. Furthermore, misinformation about cancer is widespread on online platforms such as Google and YouTube, exacerbating these challenges [3].

AI-driven chatbots have gained increasing importance by offering 24/7 assistance and personalized information to patients navigating cancer care [4], [5]. These tools can enhance patient engagement, provide emotional support, and help alleviate the workload of healthcare professionals, offering a more robust and accessible approach to cancer care [6]. Conversational AI chatbots have demonstrated the potential to provide patients with responses that are as accurate, empathetic, and readable as those provided by healthcare professionals [7]. Expert AI chatbots trained on extensive medical text corpora can serve as point-of-care electronic tools, offering essential information to vulnerable populations while also providing emotional support to cancer patients, thus improving oncology care. Additionally, chatbots have been increasingly used in applications such as cancer screening, prevention, risk stratification, treatment, monitoring, and management. Studies have shown high user satisfaction with these tools, as they enhance patient-centered communication, make cancer-related information more accessible, and improve access to care [7].

The rise of large language models (LLMs) has further advanced the capabilities of chatbots, enabling them to understand and generate human-like responses [7]. LLMs are a subset of artificial intelligence that relies on deep learning techniques, particularly transformers, to process and generate human language. These models are trained on massive datasets that include text from books, research papers, websites, and other textual resources. By leveraging this vast corpus of information, LLMs can predict and generate coherent, contextually relevant responses to a wide variety of inputs. LLMs like GPT-4 [8] and others operate through billions of parameters, which help them to capture nuances in language, including grammar, context, and meaning. When a user inputs a question or statement, the model analyzes the sequence of words and predicts the next word or phrase, building an informed response that often mirrors human conversation. These advancements in natural language processing (NLP) have significantly improved the effectiveness of chatbots in healthcare, enabling them to engage in more meaningful and empathetic interactions.

However, LLMs face significant challenges, including knowledge cutoffs and the potential for generating hallucinations that are incorrect or misleading [9]. These limitations can undermine trust and hinder the effectiveness of chatbots in delivering reliable information, especially in critical areas like oncology. To overcome these challenges, Retrieval-Augmented Generation (RAG) models offer a promising solution. By integrating sophisticated retrieval mechanisms with the generative capabilities of LLMs, RAG ensures that chatbot responses are both accurate and contextually relevant. This approach not only mitigates the risk of hallucinations but also bridges the gap created by knowledge cutoffs, providing access to the latest research and

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clinical guidelines. A RAG-based cancer chatbot can effectively address the diverse needs of cancer patients, offering tailored support across various types of cancer. This enhances patient education, supports informed decision-making, and ultimately contributes to improved health outcomes. Additionally, it alleviates the information burden on healthcare providers, allowing them to focus more on direct patient care.

II. Literature Review

The AI chatbot has immensely revolutionized the health sector, particularly in oncology. These digital tools seek to offer timely information and support to patients and increase their understanding of cancer-related issues. There is a narrative review by Piau et al. (2023) citing that chatbots have been used in many applications such as cancer screening and prevention, risk stratification, treatment management, and monitoring. Out of the 21 assessed chatbots, the majority of them indicated high patient satisfaction and positively impacted the care's efficiency compared to conventional practices. In particular, 9 of the 12 studies mentioned that the outcome of oncological care was positive through the use of chatbots [10]. Nevertheless, AI chatbots have some limitations. Some of the major restrictions of the chatbots based on AI include dependence on pre-programmed response patterns and lack of substantial contextual perceptions. This often results in too technical terms that will only confuse patients rather than help them. A review of the effectiveness of several AI chatbots on the quality of their response reported that although they had correctly provided information on the most common cancers, many of the responses were unintelligible by laypersons. In addition, such chatbots are vulnerable to inaccuracy in their training data which could be outdated or even biased, leading to providing the wrong information that has adverse effects on patient care [11], [12].

Advanced natural language processing, like more improved chatbots, paved the way to finally understanding and even responding to the demands of a patient. NLP-based chatbots possess very promising positions over simple AI systems in the sense that they allow for very personalized interaction. For instance, performance and comparison evaluation of three NLP-based chatbots were conducted against licensed physicians, and the chatbots were concluded to have outperformed human specialists in empathy and readability when answering questions related to cancer [13]. The result indicates that the mean empathy scores of the best-performing chatbot were significantly higher than those of human physicians, which may indicate that NLP improves patients' engagement and satisfaction. Despite this, NLP models have their very own limitations. Although they improve the quality of interaction, they tend to generate vague or ambiguous responses, which might confuse users. Additionally, research studies have indicated that systems based on NLP often make incorrect recommendations or are contrary to clinical guidelines. For example, one study found that a third of treatment recommendations from an NLP chatbot did not follow NCCN guidelines [4]. This inconsistency indicates that these models need to be continuously validated and updated to ensure they stay reliable.

The second major landmark in chatbot technology applied to health is Large Language Models. They are LLMs, the likes of ChatGPT, which can give meaningful coherent responses in some context to vast datasets. Because they can imitate human speech, they tend to engage patients more than the traditional model would. However, while LLMs can pump out abundant detail quickly, new problems were also introduced. Indeed, it has been documented that LLMs can produce "hallucinations," which involve generating entirely incorrect information or recommending interventions not supported by clinical guidelines; this happened in approximately 12.5% of cases during evaluations[14]. Further, LLMs may have difficulties handling nuanced medical decision-making because of their generalized training data.

After LLMs, there is an increasing interest in Retrieval-Augmented Generation models for applications in healthcare, since these models seemingly address some of the limitations of Large Language Models (LLMs). It's a combination of traditional Information Retrieval and generative capabilities that empower chatbots to give accurate, timely responses with real-time integration of verified data from medical databases [13]. The particular approach has been known to reduce diagnostic errors in health care by up to significant numbers, thus providing more precise treatment recommendations and outcomes for the patients [15]. For example, some recent implementations, as witnessed by the use of technology RAG by Apollo 24\7 in confluence with Google's MedPaLM [16], came to prove that RAG does its job effectively in bringing clinicians in real-time access to de-identified patient data as well as new medical research [13]. Of course, it's all about the models of RAG that can automate various administrative tasks such as scheduling patient visits, which made their operational efficiency better. With the synthesis of the relevant information from vast medical databases, RAG deepens the understanding of the patients as well as facilitates their cooperation-the ideal chain of communication between health care providers and patients [15]. As RAG technology evolves, it will change the medical world map in accessing and using medical information for the betterment of both patient care and operational workflows [17].

While substantial advancements have been made in elevating cancer care through enhanced patient engagement and access using AI-powered chatbots, concerns about their accuracy and reliability remain paramount. The transition from basic AI systems, then to NLP-based models, and now LLMs indicates unwavering attempts to strengthen the effectiveness of these tools in health care. However, there is still further work that needs to be done to overcome the current limitations of such technologies so they can be used accurately by patients searching for information about cancer.

III. Methodology

A. Data Collection, Chunking and Embeddings

The process for gathering data for CancerBot, a cancer-specific Retrieval-Augmented Generation system, was exhaustive and included drawing information from both medical and academic sources. Using an API key, we searched PubMed for papers with "cancer" as the main keyword. This led to the downloading of 100 full-text PDFs covering a wide range of topics, including cancer diagnosis, treatment strategies, new therapy advancements, and ongoing clinical trials. Apart from these papers, we obtained seven books from a Google search on the topic of cancer that dealt with treatment protocols, preventive measures, patient care, and educational resources for both patients and healthcare providers. The inclusion of book materials with the journal

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articles helped to provide a more comprehensive resource for the system.

After collecting the data, the next phase involved preparing it for CancerBot's RAG system. To manage the text from these papers and books, we used LangChain's ReadTheDocsLoader.[1], a tool designed for working with structured documents. Given the complexity and length of the texts, we applied a RecursiveCharacterTextSplitter.[2] To break the documents into chunks of 800 characters each, ensuring that the chunks were manageable while keeping the text coherent. Then, the sentence-transformers/all-mpnet-base-v2 model to create embeddings of the text was used. Embeddings allow the model to find contextually similar information, thus providing more accurate and meaningful responses. This is particularly important in the medical domain, where scientific literature is often dense, and understanding the context behind specific terms or treatments can drastically affect the quality of the information provided.

The sentence Transformer [18] model is well-known for generating high-quality, context-aware embeddings, which allowed CancerBot to retrieve the most relevant sections of the text based on user queries. This model is powered by pre-trained nuanced relationships between words, phrases, and sentences. These conditions make CancerBot achieve its real-time accuracy with answer-providing performance within areas like cancer treatment, research, and patient education.

B. *Vector Database*

Pinecone Vector Database[3] Was employed to efficiently manage the storage and retrieval of embeddings within the CancerBot system (Fig. 1). Pinecone is perfectly suitable for our use cases since it allows quick similarity search across large datasets. This follows the architecture of CancerBot wherein the embeddings are vital for associating user queries with pieces of cancer literature.

For the database configuration, an index in Pinecone with a dimensionality of 384 was created, reflecting the dimensions of the embeddings generated by the `sentence-transformers/all-mpnet-base-v2` model [19]. Cosine similarity as the metric for this index which can be calculated as in (1), was employed. Cosine similarity is the effective measure of the angular distance between vectors and hence is the ideal measure that should be applied for text embedding semantic content comparison. The initialization process included checking whether the index existed and, if not, then it created it. The system then monitored the index until it was fully initialized and ready for use.

$$\cos\theta = \frac{A.B}{||A||\,||B||} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}} \qquad (1)$$

Once the index was prepared, we proceeded to upsert the embeddings along with their associated metadata, which included the specific text chunks they represented. Each embedding was given an ID to track and retrieve precisely. The upsert process assures that while inserting and updating the data, there would be no duplication in such a manner that the information could be retrieved in real-time. Thus, the RAG-based system of the structure of CancerBot is capable of searching the vectorized representation of research papers and books to provide information on semantic similarity about query embeddings.

C. *Large Language Model for Generation*

This study uses the Llama-2-13b model which is part of the Llama-2 [20] family of pre-trained generative text models created by Meta and released in July 2023. They use an auto-regressive approach - which sequentially generates text based on the context provided to the model. We use the Llama-2-13b model since it is capable of handling complex linguistics since it has been pre-trained on a vast corpus of about 2 trillion tokens of text data and supports longer context lengths of up to 4096 tokens. For dialogue cases, the model has been fine-tuned in a supervised manner using labeled data. The model is a great resource to use for the use case of this study since it is built to balance both helpfulness and safety in its responses. In most cases, the Llama-2-13b model has been found to outperform most of its closed-source competitors such as ChatGPT [20].

The model is built on the powerful transformer Architecture model. The transformer model was introduced in 2017 and relies on self-attention techniques to capture contextual information from the input text. It consists of an encoder-decoder structure, where the encoder processes input tokens and the decoder generates output tokens. The multi-head attention mechanism allows the model to attend to different parts of the input sequence simultaneously, enhancing its ability to learn complex patterns [20].

The LLaMA-2 13B model was optimized for deployment on limited-memory hardware using quantization techniques. To minimize model size, quantization converts high-precision weights and activations into lower-precision formats retaining performance. By implementing this strategy, large models can be effectively run on GPUs with limited memory, expanding their accessibility for various applications with reduced computational requirements. The model was loaded with 4-bit precision to prioritize both memory efficiency and computational accuracy. The chosen quantization method maximizes performance with minimal effect on model accuracy.

[1] https://python.langchain.com/docs/integrations/document_loaders/readthedocs_documentation/

[2] https://api.python.langchain.com/en/latest/character/langchain_text_splitters.character.RecursiveCharacterTextSplitter.html

[3] https://www.pinecone.io/

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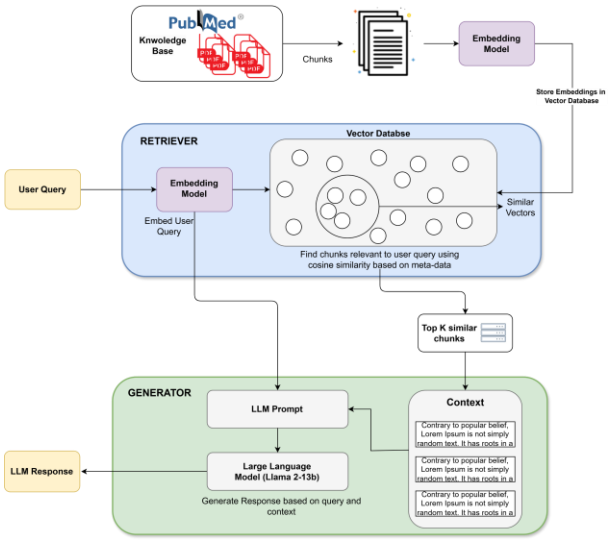


Fig. 1. Architecture of the Proposed RAG-powered CancerBot

Double quantization and lower-precision computation were employed to minimize memory and enhance speed. The LLaMA-2 13B model ran effectively on available GPU resources with this configuration, proving that large models can be deployed efficiently on constrained hardware.

D. *RAG-based Response Generation*

The retrieval-augmented generation improves the correctness of responses by combining large language models with external data sources. The RAG draws real-time factual data into its context window without having to retrain the LLM for efficient responses. It accesses this database after receiving an inquiry to gather the necessary context for the LLM's limited knowledge. After that, the LLM generates a reply based on current data. This method goes beyond the LLM as it provides context-specific resolutions.

The Knowledge Base of the cancer chatbot will be enriched using RAG to extract information from related literature. Retrieving relevant parts from the vector database deepens the context of responses to user inquiries. RAG equips the chatbot with real-time medical knowledge. RAG allows the chatbot to receive real-time updates, eliminating the need for costly LLM retraining. With new cancer data, the vector database effortlessly expands, improving the chatbot's cancer-related knowledge. By employing RAG approaches, we harness the reasoning ability and factual knowledge of a large language model, which is relationally stored and dynamic. Due to the synergy between the chatbot and the knowledge base, it delivers more elaborate responses based on current information. RAG is crucial for enabling the chatbot to store accurate and contextually relevant cancer information.

IV. Results

In this study, Gradio was utilized to develop a web-based application interface, hosted locally, for interacting with an RAG system that combines large language models LLMs with a vector database. The design of the interface created using Gradio is presented in Fig. 2, which illustrates how users interact with the system. We employed this interface to ask a series of questions to the LLM-based system, focusing on cancer-related queries. The system's responses were generated based on relevant context extracted through a similarity search from a vector database that had been populated with cancer-related documents, research articles, and medical literature.

The system generated answers that were accurate and rooted in the provided dataset by retrieving the most contextually relevant information. Fig. 3 shows how this process works, presenting both the questions posed to the system and the corresponding responses based on the retrieved context. Several questions from research papers and medical documents were selected manually and used to assess the system's performance. Among all questions, one was the question "What is the Patient Experience of Radiation Therapy for Cancer Treatment?" Here, since this model pulled out context-relevant information from the database, therefore it came up with correct response as demonstrated in Fig. 3(a). Now it confirmed the model capability of retrieving contextually relevant information relating to oxidative stress—the vital component of the disease and also could give thoughtful response for the given context.

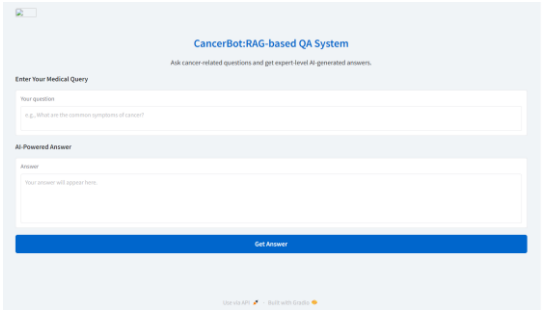
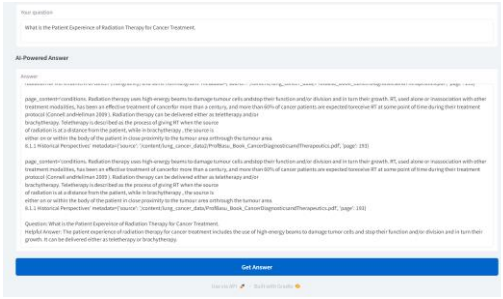
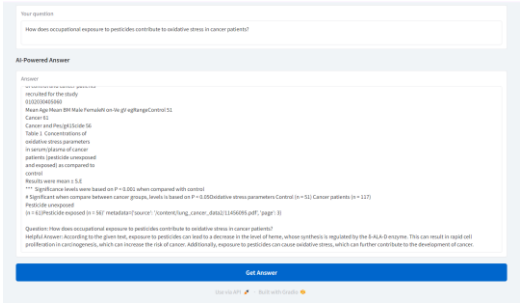


Fig. 2. User Interface of CancerBot



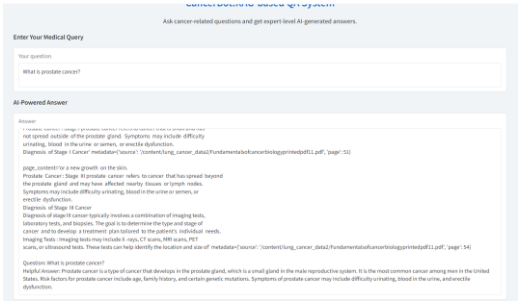
(a)



(b)

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(c)

Fig. 3. Responses by CancerBot along with retrieved context (a) What is the patient experience of radiation therapy for cancer treatment? (b) How does occupational exposure to pesticides contribute to oxidative stress in cancer patients? (c) What is prostate cancer?

In another example, the question, "How does occupational exposure to pesticides contribute to oxidative stress in cancer patients?" was posed (Figure 3 (b)). An additional instruction was given to the system to answer based strictly on the retrieved context and, if no relevant information was found, to reply, "I don't know." This instruction was implemented to mitigate the risk of hallucinations, an issue common in LLMs where they generate false or misleading information. Figure 4 illustrates how the system effectively handled this instruction, providing accurate responses based only on the available data, thereby preventing hallucinations. To test it, a control question that was unrelated to the cancer context was asked: "Who is Barack Obama?" Since the vector database did not contain any information related to Barack Obama, the system correctly responded with " can not provide information about him". This highlights the effectiveness of the RAG system in adhering to the provided context and only generating responses when relevant information is available.

After building and manually testing the RAG pipelines, we created a synthetic testing dataset along with ground truth using GPT-4. This dataset was carefully structured with headers like question, answer, context, and ground truth to ensure a thorough evaluation. To assess the performance of the RAG system, we used the Retrieval-Augmented Generation Assessments (RAGAs) [21]. The evaluation focused on several key metrics: faithfulness, context relevancy, and response relevancy. Faithfulness ensures that the response generated is contextual to prevent hallucinations and to provide factual consistency, whereas relevancy of the response addresses how well the answer has been generated about a specific question. Context Relevance determines whether or not the retrieved context is concise and irrelevant-free to maximize LLM's performance.

We tested both the LLaMA-2 model and the RAG-enhanced LLaMA-2 model by asking them the same set of cancer-related questions. The results, shown in Table 1, clearly demonstrate that the CancerBot, powered by RAG, performed significantly better than the standard LLaMA-2 13B model when handling these cancer-specific queries.

This shows that the CancerBot's ability to pull relevant information and provide accurate, contextually rich answers is much stronger due to the RAG system, especially in the complex and sensitive field of healthcare.

TABLE I. COMPARATIVE RAGAS RESULTS

Model	Evaluation Metrics		
	Faithfulness	Context Relevancy	Response Relevancy
Llama 2-13B	0.54	0.41	0.34
RAG-powered Llama 2-13b	0.83	0.72	0.79

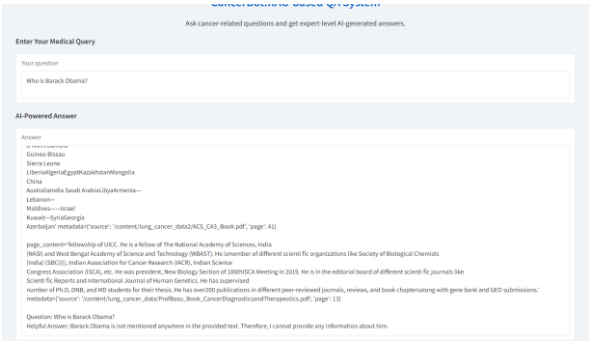


Fig. 4. CancerBot dealing with hallucinations

However, despite its brilliant prospects, there are a couple of scopes where improvement is needed in order to take it to the next level. Currently, it relies on cosine similarity for retrieving relevant information. Although this works very well in most cases, this methodology might not be useful in some instances in bringing the outcomes closest to perfection or even contextual accuracy [22]. The size of the corpus that it is currently using is also pretty small. Though adequate for its current scope, it limits the range of cancers and medical concerns it can address entirely.

V. CONCLSUION AND FUTURE DIRECTIONS

The development and evaluation of the RAG-based CancerBot highlight important progress in how AI can help retrieve and generate accurate, contextually relevant information for cancer-related questions. CancerBot became a better model combined with a vector database as opposed to the standard LLaMA-2 model. Answers became clearer and more accurate, bringing down the possibility of spreading misinformation. Additionally, the use of Gradio for the interface made the system more accessible, offering a user-friendly and intuitive way for people to interact with the tool. Future development would be to expand the corpus to address more diverse kinds of cancers and even more medical concerns. A further step would be to consider better retrieval methods than cosine similarities to extend relevance and precision in searching information. We will also experiment with different open-source models and embedding techniques in our attempt to make the system even more adaptive and efficient. Another interesting possibility would be its integration into wearable technology like smart glasses, which makes it more portable and easy to access. In conclusion, this paper demonstrates the possibilities of Retrieval-Augmented Generation systems in truly impacting health care by producing reliable information that is based on actual evidence.

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